

HARTFORD-BRAINARD, CT, USA

WMO: 725087

Lat: 41.736N

Lon: 72.651W

Elev: 6

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14752

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.2	-22.7	0.5	-11.1	-20.3	0.6	-9.3	10.1	0.4	9.0	-2.2	3.1	320	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	32.9	23.0	31.4	22.4	29.9	21.6	24.9	30.1	24.1	28.7	23.3	27.4	3.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.2	27.2	22.7	17.4	26.4	22.1	16.8	25.9	75.6	30.0	72.4	28.9	69.3	27.1	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.6	6.9	-17.4	35.7	2.7	1.9	-19.3	37.0	-20.9	38.1	-22.4	39.2	-24.3	40.6		
(5)			-18.0	26.4	2.5	1.0	-19.8	27.1	-21.3	27.7	-22.7	28.3	-24.5	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-1.5	-0.3	3.8	10.2	15.9	20.8	24.0	23.1	19.1	12.6	6.5	1.3		
	DBStd	9.89	5.61	5.12	4.93	4.44	4.01	3.53	2.85	2.94	3.75	4.25	4.52	4.80		
	HDD10.0	1314	357	290	200	49	2	0	0	0	0	22	122	273		
	HDD18.3	3051	614	522	450	247	98	16	1	2	35	184	354	528		
	CDD10.0	1813	1	1	9	55	186	324	433	407	273	102	18	3		
	CDD18.3	507	0	0	0	3	24	89	175	151	59	6	0	0		
	CDH23.3	4181	0	0	4	48	251	747	1521	1188	388	34	1	0		
	CDH26.7	1394	0	0	1	15	78	248	562	391	95	4	0	0		
(14) Wind	WSAvg	2.9	3.1	3.2	3.5	3.4	3.0	2.8	2.6	2.3	2.5	2.7	2.9	2.9		
(15) Precipitation	PrecAvg	1095	82	78	88	97	98	96	94	86	97	85	94	93		
	PrecMax	1468	325	220	209	278	281	303	204	236	249	192	201	197		
	PrecMin	722	10	10	16	31	16	37	18	7	38	11	12	15		
	PrecStd	212	65	42	46	52	57	64	45	58	60	43	49	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	16.0	22.7	28.8	32.2	33.7	35.9	34.7	32.1	26.9	21.1	17.5		
		MCWB	12.5	10.7	13.8	17.7	20.4	23.5	24.7	24.4	23.1	18.9	16.0	13.7		
	2%	DB	12.4	12.3	17.5	23.7	28.8	31.6	33.3	32.1	29.1	23.7	18.0	13.8		
		MCWB	9.6	8.0	11.0	14.5	19.2	22.2	23.3	23.2	21.1	18.4	14.5	11.4		
	5%	DB	8.9	9.3	14.0	20.7	26.2	29.5	31.6	30.4	27.3	21.8	16.3	11.4		
		MCWB	6.1	5.6	8.7	12.7	17.7	21.0	22.5	22.2	20.5	17.4	12.7	9.1		
	10%	DB	6.2	7.1	11.5	17.9	23.4	27.7	29.8	28.8	25.6	19.8	14.1	8.7		
		MCWB	3.3	3.5	7.1	11.3	16.4	20.2	22.0	21.3	19.6	15.9	10.7	6.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	12.9	15.5	18.9	22.4	24.9	26.5	25.9	24.3	21.9	17.9	15.0		
		MCDB	15.5	14.0	21.2	26.8	29.5	31.4	32.1	31.9	28.6	24.3	19.5	16.7		
	2%	WB	10.2	8.9	12.4	16.7	20.7	23.7	25.1	24.8	23.2	19.9	15.6	12.1		
		MCDB	11.6	11.1	15.7	21.9	25.9	29.2	30.4	29.4	26.7	22.3	17.4	13.4		
	5%	WB	6.4	5.8	9.8	14.5	19.3	22.7	24.3	24.0	22.3	18.2	13.7	9.1		
		MCDB	9.3	8.5	13.2	18.3	24.4	27.3	28.9	27.9	25.6	21.0	15.8	11.2		
	10%	WB	3.7	3.9	7.6	12.5	17.8	21.6	23.4	23.0	21.1	16.5	11.4	6.4		
		MCDB	5.7	6.8	10.8	16.5	21.8	25.6	27.6	26.4	24.1	19.3	13.6	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	9.5	10.7	10.9	10.3	10.0	9.8	10.2	10.0	9.0	7.8		
		MCDBR	11.0	12.0	13.8	15.3	14.7	12.9	11.7	11.3	11.6	12.2	11.8	11.0		
	5% WB	MCWBR	9.0	8.6	8.4	7.9	7.0	5.6	4.2	4.3	5.3	7.2	8.7	9.7		
		MCWBR	10.7	11.0	12.2	12.9	12.6	11.1	9.7	9.4	9.2	10.5	10.5	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.315	0.339	0.374	0.418	0.463	0.484	0.452	0.380	0.345	0.316	0.301		
		taud	2.448	2.424	2.428	2.366	2.269	2.181	2.133	2.214	2.389	2.500	2.539	2.530		
	Ebn at Noon		849	893	910	896	859	819	796	811	853	849	831	825		
		Edh at Noon	77	91	103	117	133	145	151	135	106	85	70	66		
(44) All-Sky Solar Radiation	RadAvg		1.84	2.69	3.68	4.75	5.30	5.76	5.92	5.26	4.34	2.87	2.01	1.51		
	RadStd		0.15	0.24	0.28	0.51	0.54	0.57	0.34	0.29	0.34	0.28	0.16	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102	+53

Nomenclature: See separate page